11.1 Calculate and use conditional probabilities to include Bayes' theorem for up to three events, including the use of tree diagrams.	

12 Probability distributions

Subject content	Additional information
12.1 Know the use and validity of distributions which could be appropriate in a particular real world situation: binomial, normal, Poisson and exponential.	
12.2 Evaluate the mean and variance of linear combinations of independent random variables through knowledge that if X_i are independently distributed $\left(\mu_i, \sigma_i^2\right)$ then $\sum a_i X_i$ is distributed $\left(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2\right)$.	Students should know that $\mathrm{E}(aX \pm bY) = a\mathrm{E}(X) \pm b\mathrm{E}(Y)$